INFO 440

Wikipedia Edit Histories: How is Truth or Belief “Drafted”?



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WIKIPEDI

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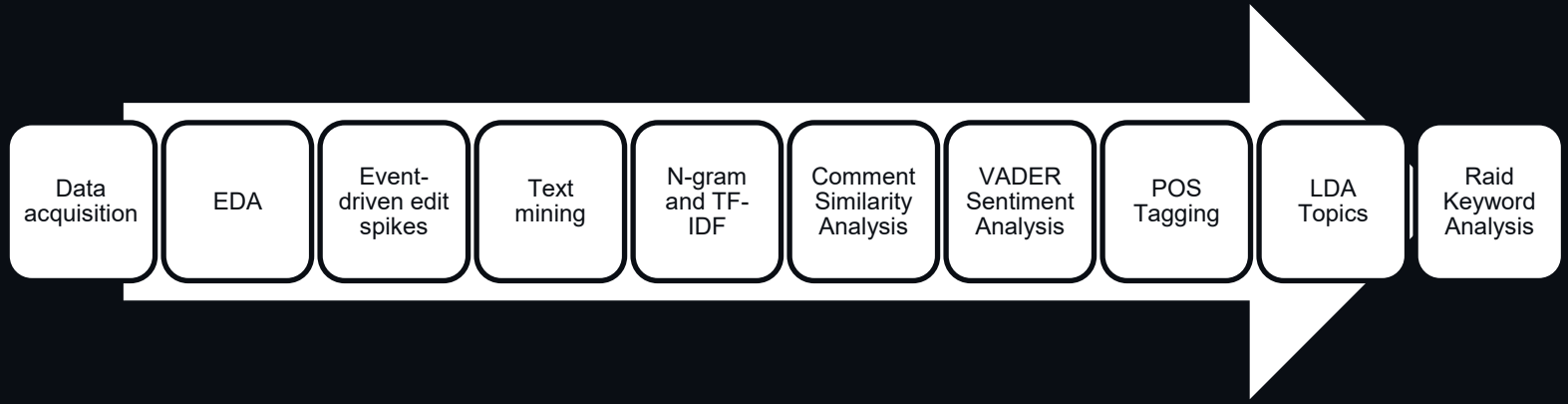
Overview

Overview - Motivation

What does Wikipedia's edit history tell us about how communities find the truth?

Overview – Data and Roadmap

- **Source:** Wikipedia API
- **Timeframe:** Full historical data to identify peak traffic months
- **Articles Studied:**
  - *Socio-Political:* 2026 Iran War, 2026 Lebanon War, Donald Trump, Joe Biden, Hantavirus
  - *Scientific/Cultural:* Artemis II, 2026 in Science, 2026 in Film



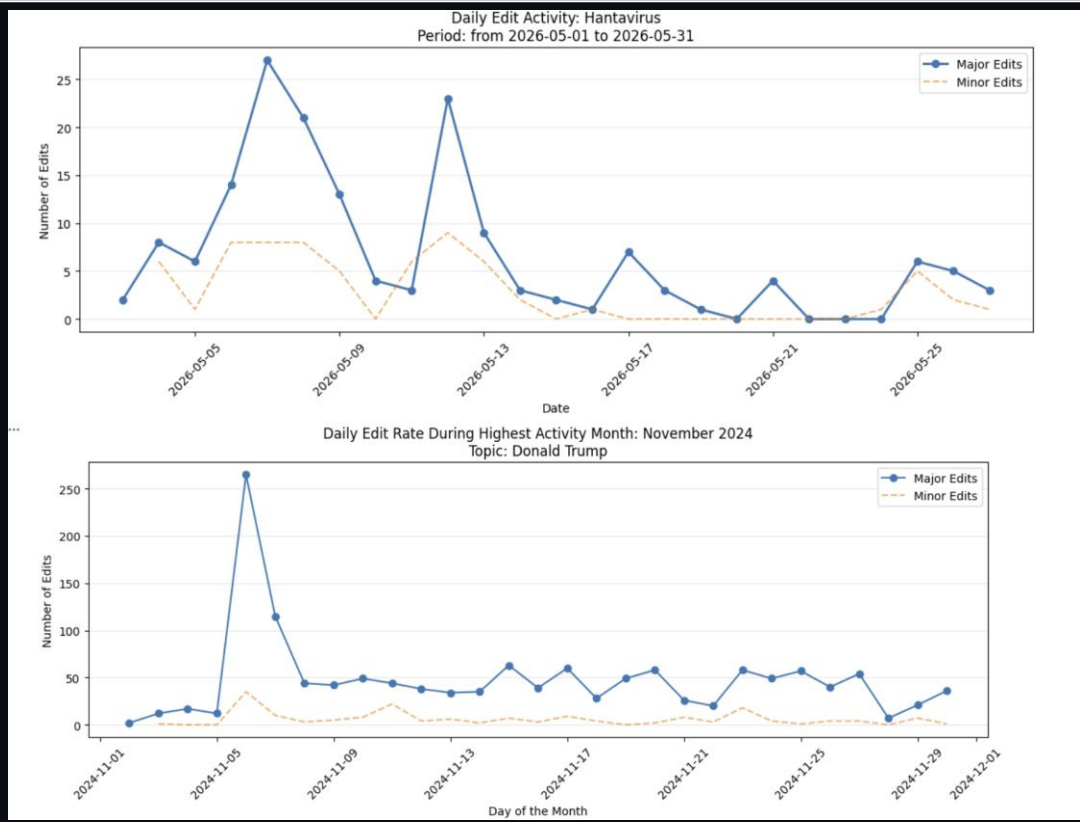


02

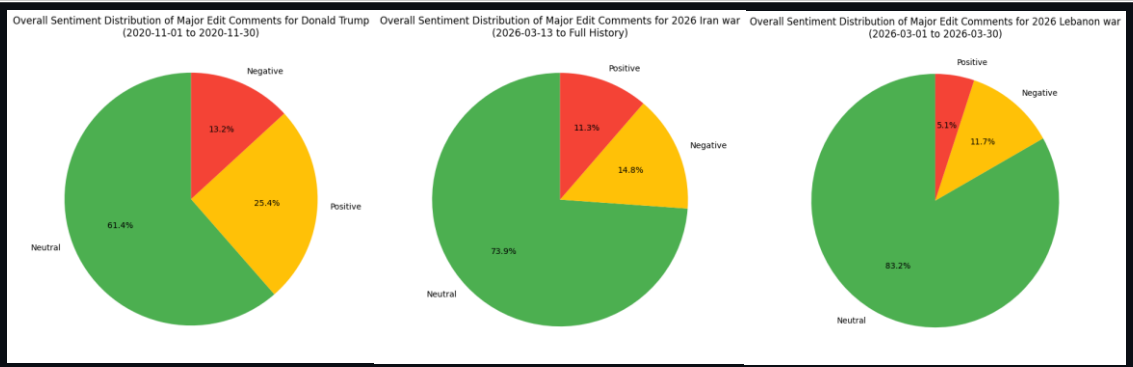
Models & Findings

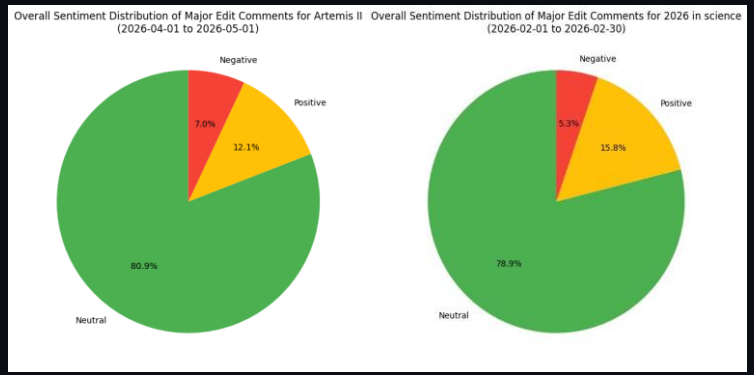
Methods – Edit Spikes

- Edit rate analysis in peak months
  - Examined major edit rates
- Identified spikes in visual analysis
  - Hypothesized that secondary spikes indicated controversial topics/ vandalism



Methods – VADER Sentiment

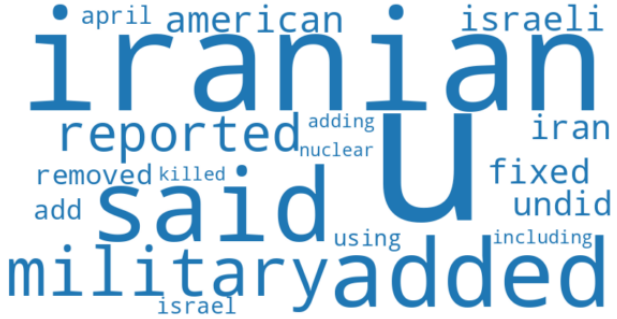




- Political articles show the highest negative sentiment
- Scientific topics are more objective and procedural
- Although being political/ controversial, 2026 Lebanon war graph looks different

Methods – POS, LDA

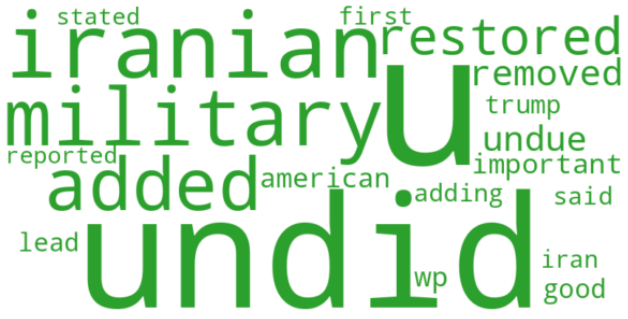
Top Words: Negative



Top Words: Neutral



Top Words: Positive

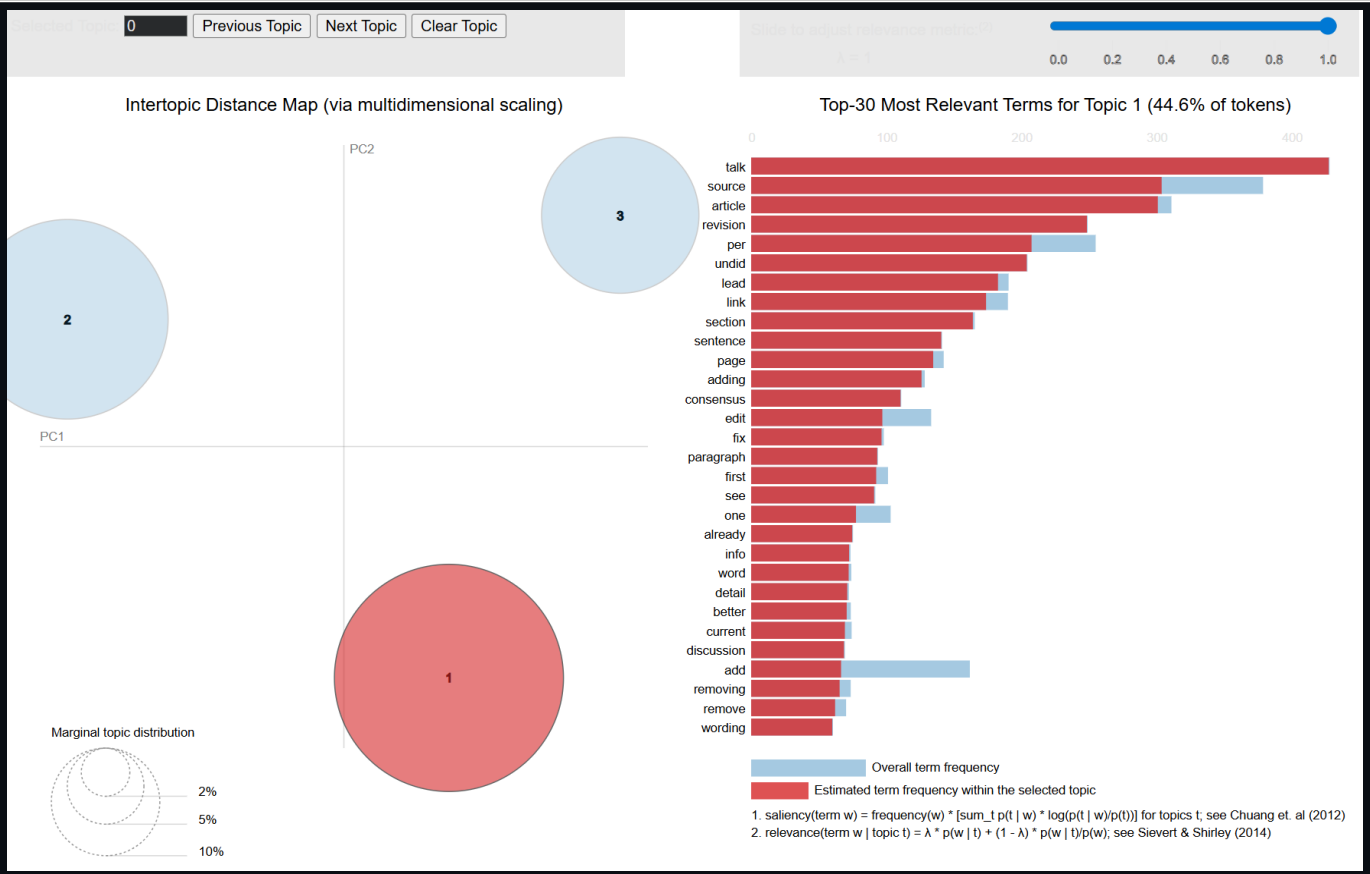



```python
def get_pos_tags(text):
    tokens = nltk.word_tokenize(str(text))
    tagged_text = nltk.pos_tag(tokens)
    return tagged_text

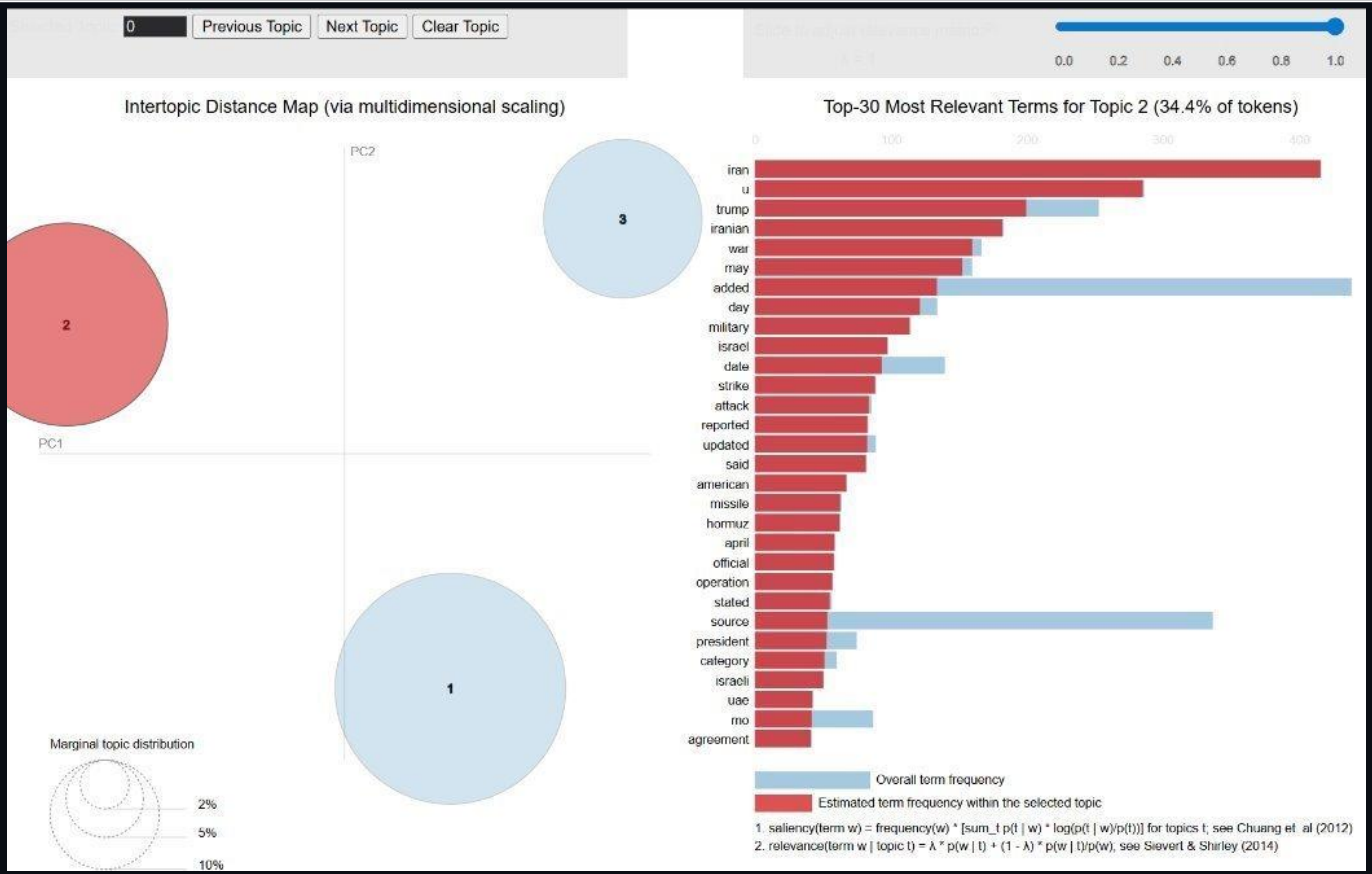

def get_words_by_pos(pos_tags, tag_prefixes):
    words = []
    for word, tag in pos_tags:
        if any(tag.startswith(prefix) for prefix in tag_prefixes):
            words.append(word)
    return words
```

- Use POS for top 20 words across negative, neutral, and positive sentiment
  - Similar words like add/adding/added, fixed, removed, undid
- Negative words aren't uniquely raid keywords; just concentrated in controversial articles
  - Use LDA to see if we can understand more of what is being commented

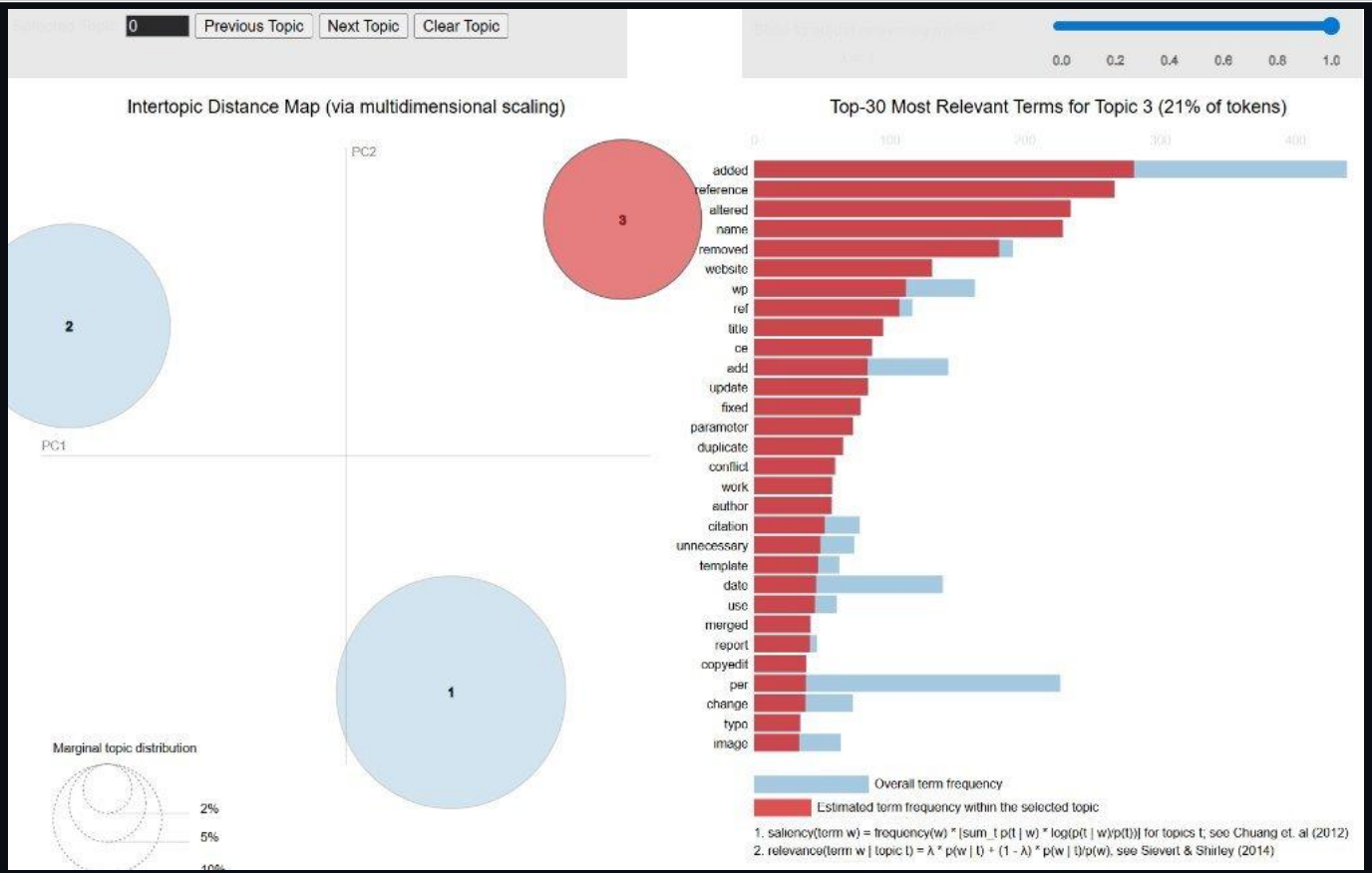
Methods – POS, LDA



Methods – POS, LDA

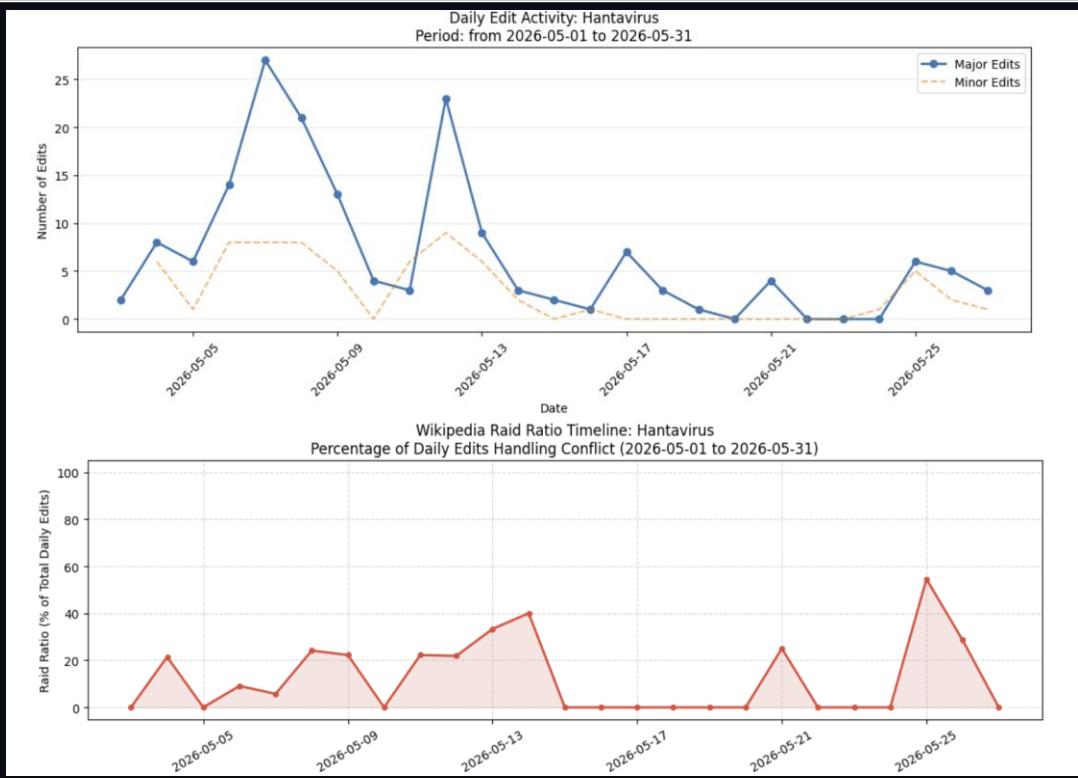


Methods – POS, LDA



Methods – Raid Keyword Analysis

- Found words that indicate vandalism
  - revert, undid, rvv, vandal, hoax, fake, disinfo, false, attack
- Ran analysis on top months in edit activity for vandalism
  - Controversial usually had higher top months
- Ran analysis of percentage of vandalism over time
  - Saw that vandlism was most present in edit rates spikes
  - Especially so in secondary spikes



Findings and Conclusion

Findings

- Controversial articles had higher negative comments
- Negative comment sentiment sometimes correlated with edit rates
- Raid analysis found that edit rates and raid words activity rose together in secondary spikes

Conclusions

- During initial spikes, articles are more likely to be focused on truth finding
- However, secondary spikes indicate vandalism
- Thus, we concluded that a secondary spike after a first initial spike should indicate vandalism



03

Challenges & Outlook

Challenges

Data Size

- Was easy to go beyond scope (forget that edit rates is the core/ tie back)
- Had to scrap edit content analysis given said content was pages-long
- Had to find controversial articles through research and assumptions
    - Had to determine the articles' viability post-analysis

Analysis

- Had to determine raid words after:
    - Failing to find unique keywords in top most negative words
    - Failing to find a unique topic that was relevant to raids via LDA

Outlook

- Learned how to work with an API to pull historical data
- Applied various data analytic practices to a large temporal data set
- Iterated on own ideas to eventually find a combination of tools that helped form a conclusion

Thank You!

Do you have any questions?

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WIKIPEDI

01

Data

Overview - Data

Wikipedia Edits

- Created a function where entering a title and time period retrieves:
  - Date edited
  - Minor or major change
  - Edit comment
  - Edit content (include_content = T/F)
- Allows for flexibility in choosing what articles to grab information from

```python
def get_refined_history(title, start_date=None, end_date=None, include_content=False):
    url = "https://en.wikipedia.org/w/api.php"

    headers = {
        'User-Agent': 'EditVisualizerBot/1.0 (cdickersondm@gmail.com)'
    }

    # Base parameters
    params = {
        "action": "query",
        "format": "json",
        "prop": "revisions",
        "titles": title,
        "rvlimit": "max",
        "rvprop": "timestamp|flags|comment",
        "rvdir": "newer"  # Set to 'newer' so we search forward in time
    }

    # Filter for actual edit text if requested
    if include_content:
        params["rvprop"] = "timestamp|flags|comment|content"
        params["rvslots"] = "main"

    # Add date filters if provided
    if start_date:
        # Wikipedia expects ISO format: 2026-01-01T00:00:00Z
        params["rvstart"] = f"{start_date}T00:00:00Z"
    if end_date:
        params["rvend"] = f"{end_date}T23:59:59Z"
```

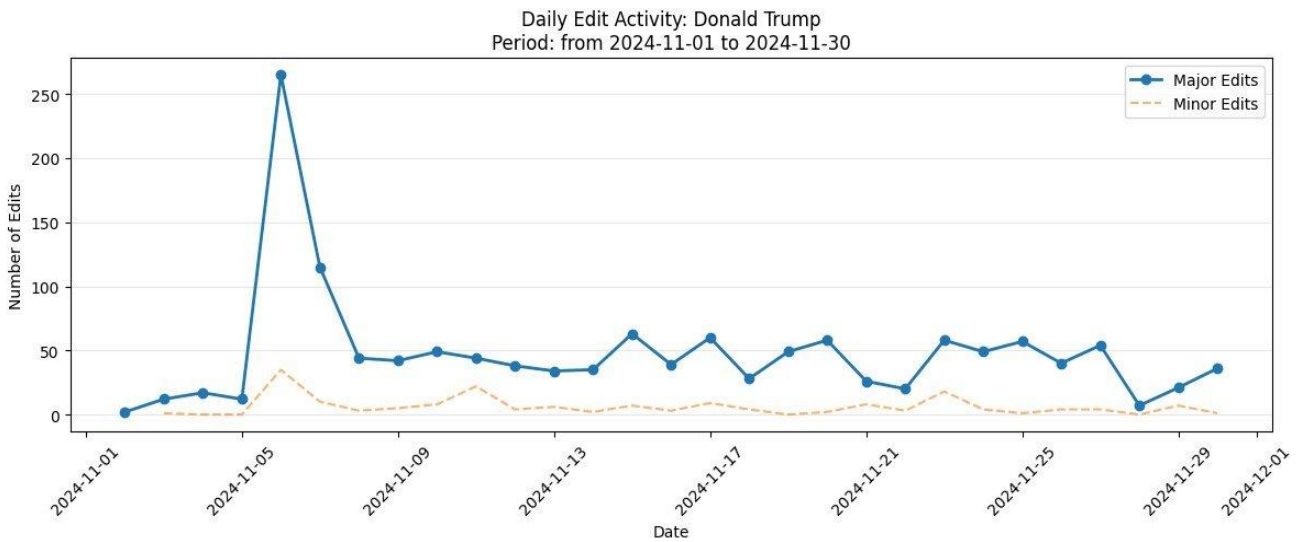
Overview - Data

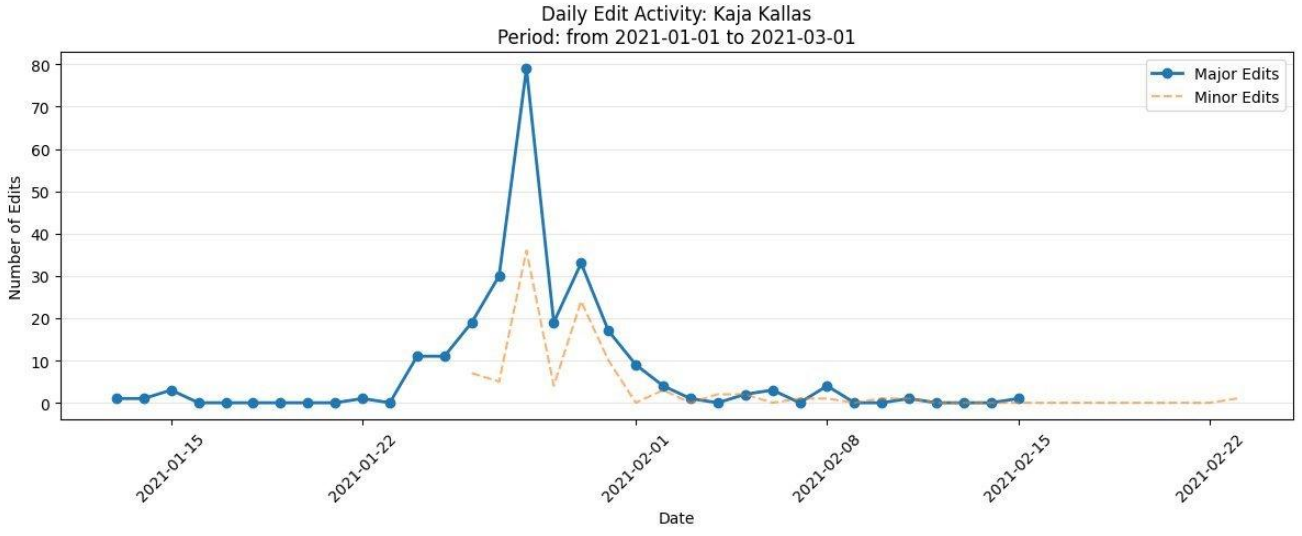
Starting Articles

- Focused on a mix of both controversial (recent and past) and non-controversial (event and discovery) articles
- Visualizes edit rates of their highest edit month, analyzing edit comments and content

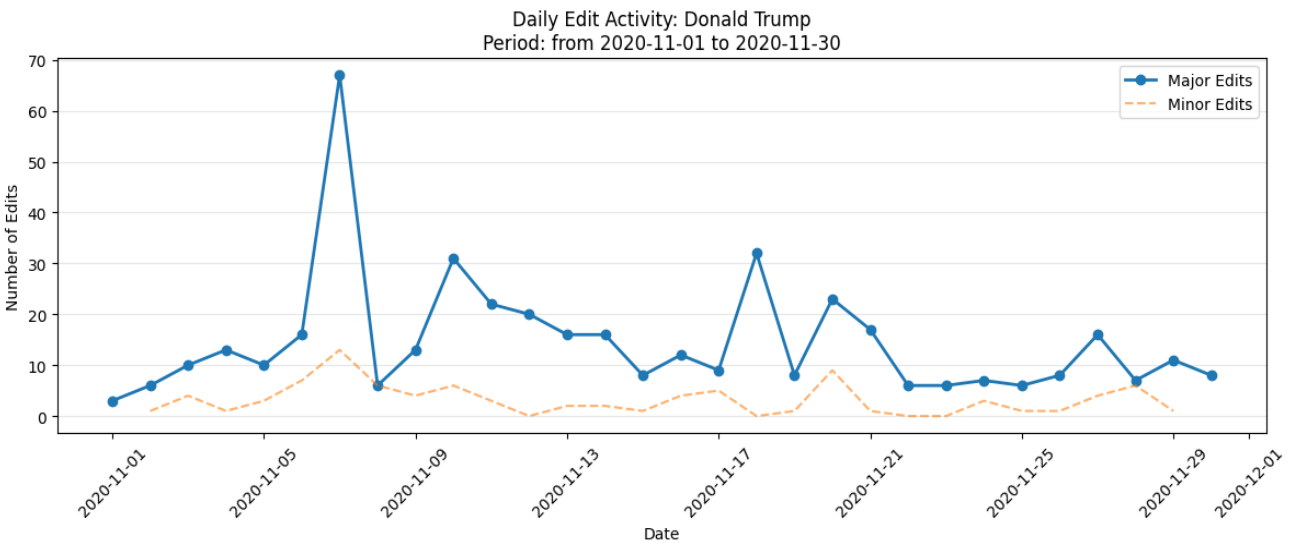
- 2026 Iran war (Socio-political, Recent);
- Donald Trump (Socio-political, Legacy);
- Artemis II (Scientific, Event-Driven);
- 2026 in science (Scientific, Discovery);

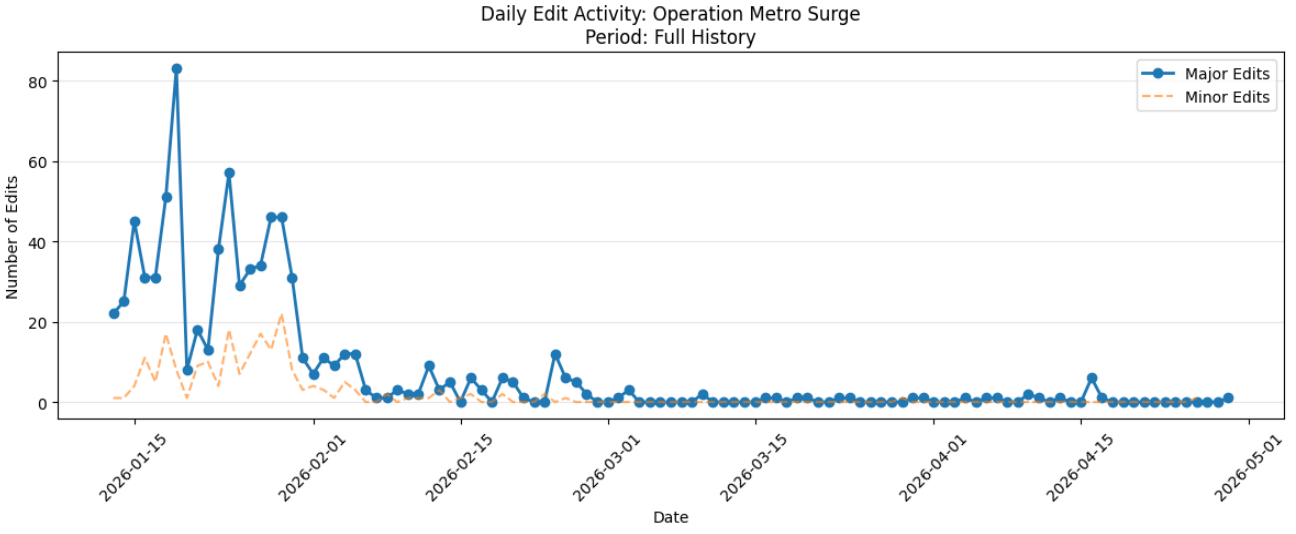
Trend found: Non-Controversial





Trend found: Controversial







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02

Model

Model

Clean Wikipedia Edit Comments

- Created a function that:
  - Convert edit comments to Unicode
    - nbsp prevents consecutive whitespace characters from collapsing into a single space
  - Remove section markers and URL
  - Convert internal links to text
    - [[LinkText]]
  - Keep letters, numbers, and spaces
  - Remove # and whitespace

```python
import re
import html

def clean_edit_comment(text):
    text = str(text)

    # Decode HTML entities twice in case comments are double-escaped
    text = html.unescape(html.unescape(text))

    # Replace non-breaking spaces with normal spaces
    text = text.replace('\xa0', ' ')

    # Remove Wikipedia edit section markers, like /* Background */
    text = re.sub(r'/\*.*?\*/', ' ', text)

    # Remove URLs
    text = re.sub(r'https?://\S+', '', text)

    # Convert internal links to display text
    text = re.sub(r'\[\[(?:[^|\]]*\|)?([^\]]+)\]\]', r'\1', text)

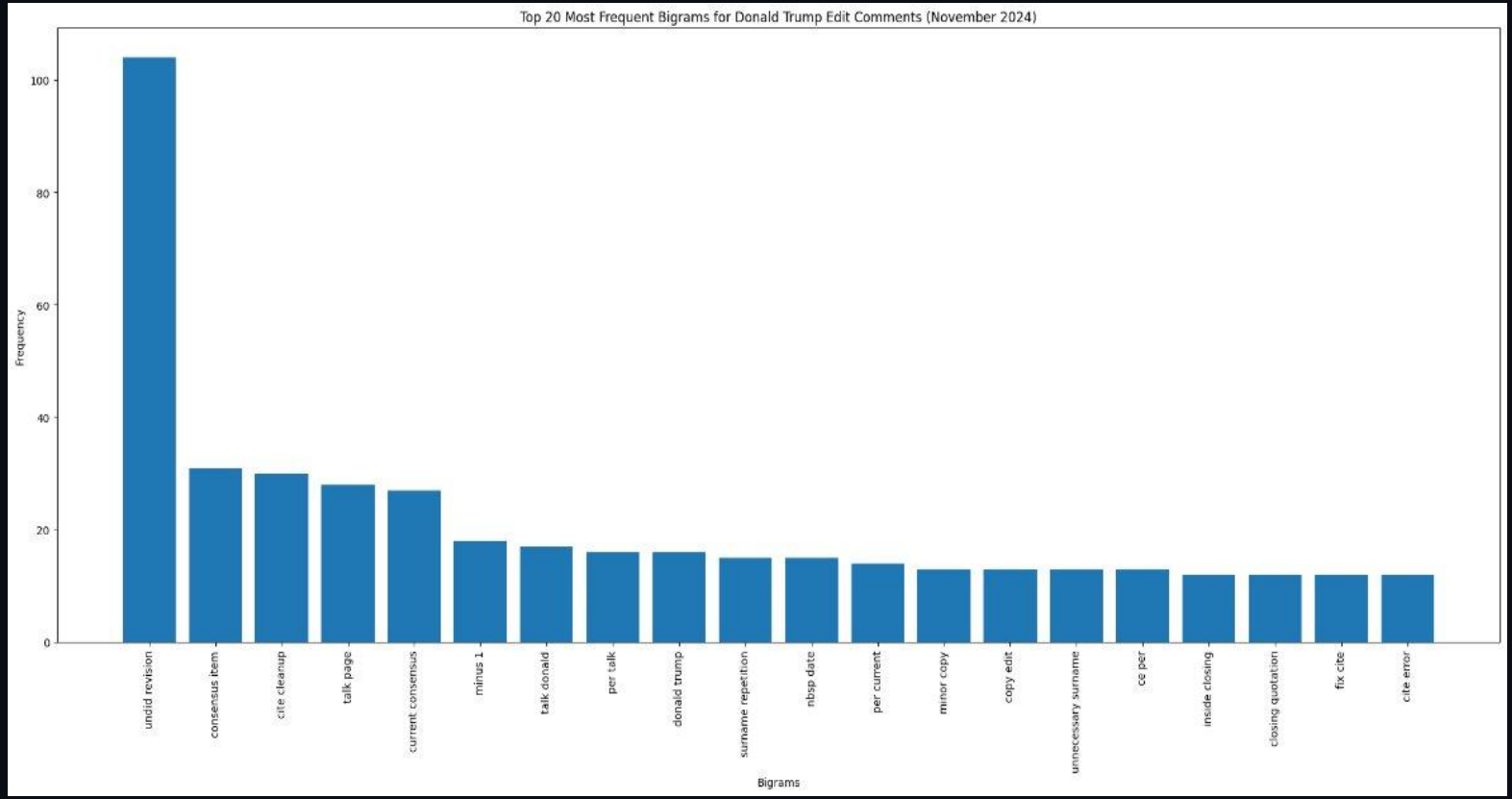
    # Remove hashtags but keep the word
    text = re.sub(r'#(\w+)', r'\1', text)

    # Keep letters, numbers, and spaces
    text = re.sub(r'[^a-zA-Z0-9\s]', ' ', text)

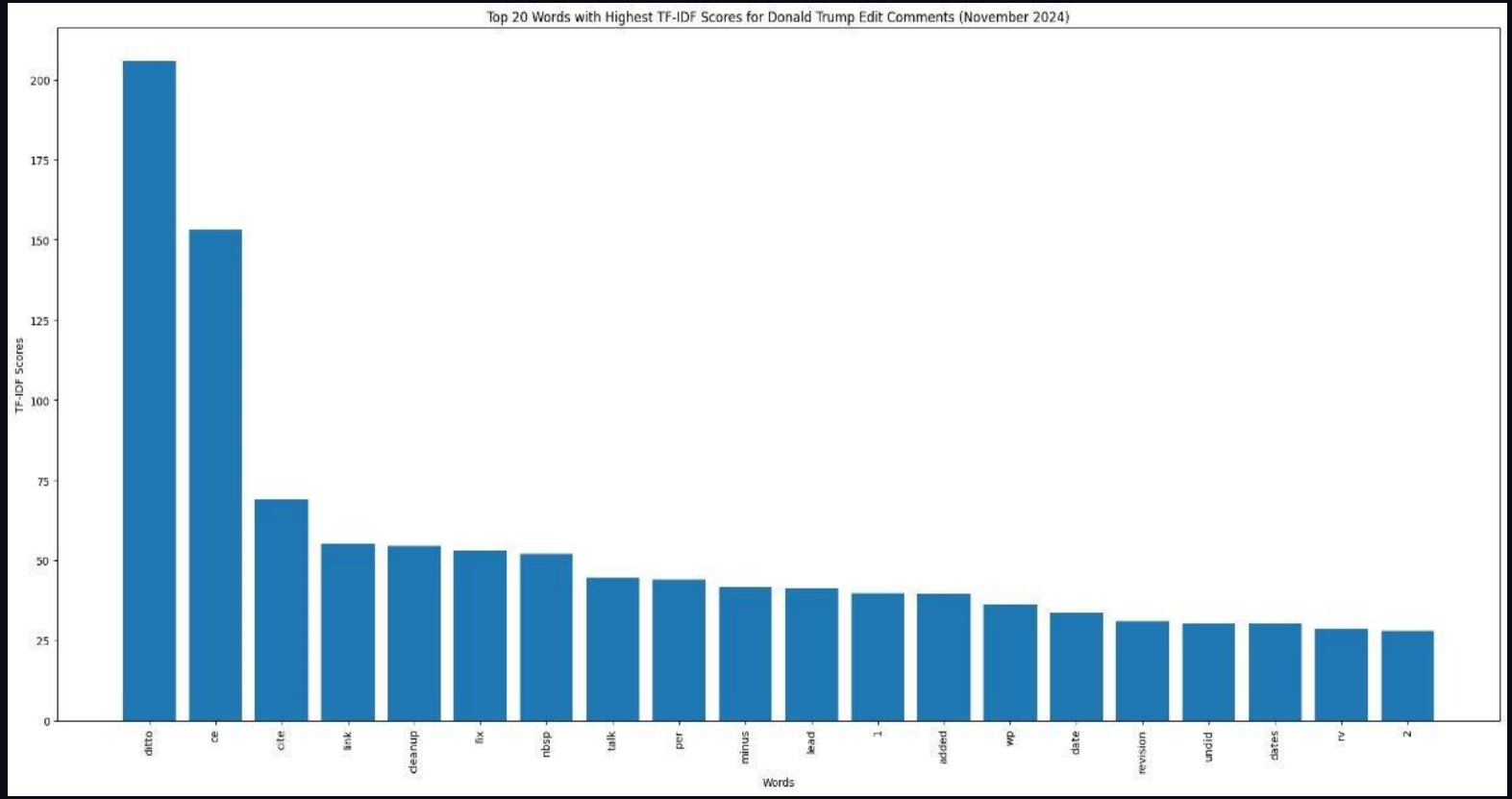
    # Clean whitespace
    text = re.sub(r'\s+', ' ', text).strip()

    return text.lower()
```

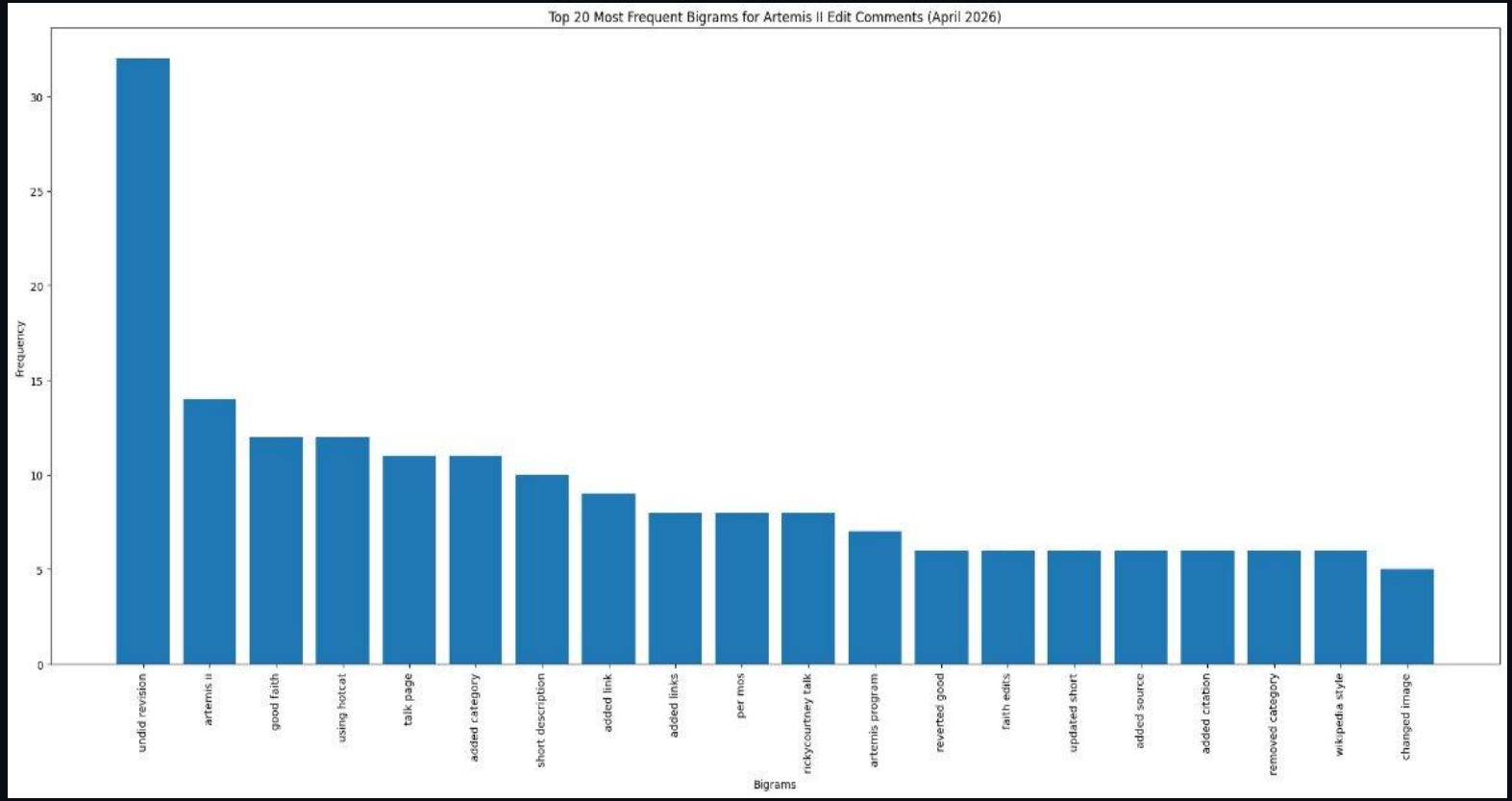
Donald Trump (Nov 2024)



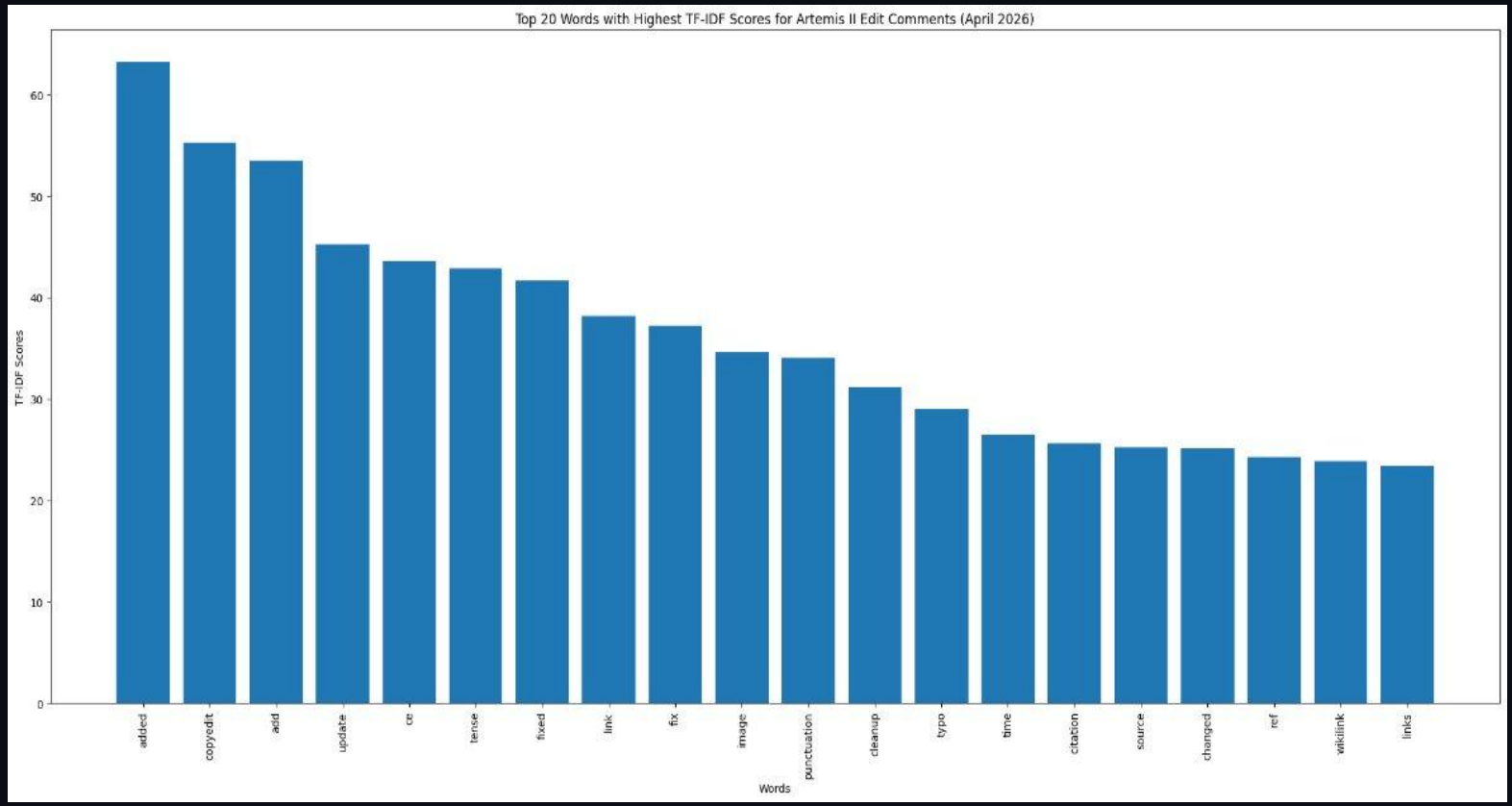
Donald Trump (Nov 2024)



Artemis II (Apr 2026)



Artemis II (Apr 2026)





03

Challenges

Challenges

Finding Articles

- Many articles are under protection, complicating edit rates
  - Had to assume what articles are 'controversial'

Data Gathering

- Edits are long and can range throughout an entire article
  - Causes long compute time due to edit length



04

Updates

Updates

Analyzing Comments and Content

- Analyze sentiment analysis further to identify any signs of 'raids'
- Find words used when 'raids' occur
- Analyze the edit rates in 'raids' to find patterns that can be generalized

Article Finding

- Use found patterns to see whether or not it is possible to identify articles that:
  - are currently 'raided'
  - have a history of 'being raided'